TEST YOUR UNDERSTANDING

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Levels 1-2: Remembering/Understanding

1. One characteristic that distinguishes a population in an extinction vortex from most other populations is that
(A) its members are rare, top-level predators.
(B) its effective population size is lower than its total population size.
(C) its genetic diversity is very low.
(D) it is not well adapted to edge conditions.

2. The main cause of the increase in the amount of CO_2 in Earth's atmosphere over the past 170 years is
(A) increased worldwide primary production.
(B) increased worldwide fertilizer production.
(C) increased infrared radiation absorption by the atmosphere.
(D) the burning of fossil fuels and deforestation.

3. What is the single greatest threat to biodiversity?
(A) overharvesting of commercially important species
(B) habitat alteration, fragmentation, and destruction
(C) introduced species that compete with native species
(D) novel pathogens

Levels 3-4: Applying/Analyzing

4. Which of these is a consequence of biological magnification?
(A) Toxic chemicals in the environment pose greater risk to top-level predators than to primary consumers.
(B) Populations of top-level predators are generally smaller than populations of primary consumers.
(C) The biomass of producers in an ecosystem is generally higher than the biomass of primary consumers.
(D) Only a small portion of the energy captured by producers is transferred to consumers.

5. Which of the following strategies would most rapidly increase the genetic diversity of a population in an extinction vortex?
(A) Establish a reserve that protects the population's habitat.
(B) Introduce new individuals transported from other populations of the same species.
(C) Sterilize the least fit individuals in the population.
(D) Control populations of the endangered population's predators and competitors.

6. Which of the following statements about protected areas established to preserve biodiversity is true?
(A) About 25% of Earth's land area is now protected.
(B) National parks are the only type of protected area.
(C) Management of a protected area does not need to be coordinated with management of the surrounding area.
(D) It is especially important to protect biodiversity hot spots.

Levels 5-6: Evaluating/Creating

7. DRAW IT Suppose you are managing a forest reserve, and one of your goals is to protect local populations of woodland birds from parasitism by the brown-headed cowbird. You know that female cowbirds don't venture more than about 100 m into a forest and that nest parasitism is reduced when woodland birds nest away from forest edges. The reserve extends about 6,000 m from east to west and 3,000 m from north to south. It is surrounded by a deforested pasture on the west, an agricultural field for 500 m in the southwest corner, and intact forest everywhere else. You must build a road, 10 m by 3,000 m, from the north to the south side of the reserve and construct a maintenance building that will take up 100 m^2. Draw a map of the reserve and explain where you would put the road and building to minimize cowbird intrusion along edges.

8. EVOLUTION CONNECTION The fossil record indicates that there have been five mass extinction events in the past 500 million years (see Concept 25.4). Many ecologists think we are on the verge of another. Briefly discuss the history of mass extinctions and the length of time it typically takes for species diversity to recover through evolution. Explain why this should motivate us to slow the loss of biodiversity today.

9. WRITE ABOUT A THEME: INTERACTIONS One factor favoring rapid population growth by an introduced species is the absence of the predators, parasites, and pathogens that controlled its population in the region where it evolved. In a short essay (100–150 words), explain how evolution by natural selection would influence the rate at which native predators, parasites, and pathogens attack an introduced species.

10. SYNTHESIZE YOUR KNOWLEDGE

Big cats, such as the Siberian tiger (*Panthera tigris altaica*), are one of the most endangered groups of mammals in the world. Based on what you've learned in this chapter, discuss some approaches you would use to help preserve them.

For selected answers, see Appendix A.

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Instructors: Questions can be assigned in Mastering Biology.